An Attention-based Unsupervised Framework for Video Anomaly Detection in Large Heterogeneous Environments

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***Abstract*—Although automated detection capabilities are frequently included in modern video anomaly detection systems, these algorithms may still encounter issues including high false-positive rates and trouble comprehending complicated circumstances. Dealing with big heterogeneous datasets that are taken in various lighting and environmental circumstances with changing resolution and quality results in obstacles for video anomaly detection systems. This paper proposes an end-to-end vision transformer (ViT)-based framework for video anomaly detection in surveillance systems. The proposed framework incorporates the ViT model with a spatio-temporal attention model augmented with Long Short-Term Memory (LSTM) layers to extract both the global context and the spatio-temporal features of the video frames. The framework is mainly designed to work with large and heterogeneous datasets that represent different environments, which remains challenging for the current video anomaly detection efforts. To assess the proposed framework, video anomaly benchmark datasets are used: the big ShanghaiTech dataset, the CUHK Avenue dataset, and the UCSD-Ped2 dataset. The proposed framework achieved AUC_ROC of 79.6%, 83.4%, and 93.8% for the three datasets, respectively. This illustrates the superior performance of the proposed framework in identifying video abnormalities compared to SOTA, especially for large heterogeneous datasets, i.e., ShanghaiTech.**

Keywords— video anomaly detection, vision transformer, surveillance systems, large datasets, unsupervised learning, spatiotemporal modeling

I. INTRODUCTION

Video anomalies are activities or occurrences that deviate from known or usual patterns or conventions in videos [1] [2]. These variations might differ significantly depending on the context of the video analysis. A surveillance system is one that observes and tracks activities and behaviors continuously. Anomalies are unusual occurrences like accidents, break-ins, or unapproved entry to restricted locations. Since anomalous events are uncommon, it could be difficult for observers to identify them in records. In addition, these surveillance systems are widely recognized for having heterogeneous recorded videos with various environmental conditions and different abnormalities types. Thus, it's crucial to provide methods for automatically recognizing these abnormal patterns. The main difficulty encountered in video abnormality detection is the variety of activities that are deemed normal in one context but abnormal in another. Running, for example, is a fairly normal and natural form of exercise, doing so inside a bank is seen as unusual or suspicious.

The increasing amount of video data produced by these systems necessitates the development of an effective and dependable anomaly detection technique, which still needs to be improved to accomplish given the dynamic and intricate nature of video footage. Conventional methods frequently depend on labor-intensive, error-prone, and inefficient manual monitoring or crude automated systems. The discrepancy in data volume between the anomaly and normal chunks make it more difficult to discover and detect abnormalities than in other video analysis techniques since abnormal actions are more uncommon than regular ones. This has raised interest in automating and enhancing the accuracy of anomaly identification in video surveillance by utilizing cutting-edge machine learning techniques, especially deep learning. Inspired by our work in [3], we introduce an unsupervised end-to-end video anomaly detection framework for surveillance system. The framework utilizes the power of the recent vision transformer (ViT) attention mechanism to capture the global context and long-range relationships within the video frame. Originating from challenges related to natural language processing (NLP), the ViT evolved as a revolutionary deep learning model [4]. Recently, ViTs have gained interest in the field of image analysis, particularly for the identification of anomalies in images. The traditional deep learning models mainly uses convolutional (Cov.) layers, but ViTs instead, utilizing the self-attention mechanisms. This mechanism enables ViTs to gather the global relationships between picture patches. In addition, a spatio-temporal module is utilized to obtain the spatio-temporal features between video frames to learn the activities and relations between objects in surveillance videos, which enable the framework to differentiate between normal and abnormal events. In this paper we propose an unsupervised video anomaly detection framework to detect abnormalities in heterogeneous and large video anomaly data that were collected under different environmental circumstances, with varying resolutions and qualities.

II. RELATED WORK

The reconstruction-based video anomaly detection methods rely on models, typically autoencoders or generative models, to

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learn how to reconstruct "normal" video frames or sequences. During inference, the model attempts to reconstruct incoming frames, and anomalies are detected when the reconstruction error is significantly higher than usual, indicating that the model is struggling to accurately reconstruct something it has not seen during training. The proposed model follows the reconstruction-based framework for detecting anomalies in videos by utilizing an attention-based encoder-decoder approach.

Hasan et al. [5] introduced a technique that learned the temporal regularities using autoencoders to identify anormal activities in long video. A fully convolutional feed-forward autoencoder was implemented to learn local features to capture regular motion patterns in video sequences. Luo et al. [6] proposed a model called "ConvLSTM-AE" for detecting anomalies in videos. The model incorporated a Convolutional LSTM (ConvLSTM) into an Auto-Encoder (AE) architecture. The appearance of each frame was encoded by a CNN and a ConvLSTM for capturing and memorizing the motion from past frames. The AE architecture was used to reconstruct frames to detect irregularities depending on a reconstruction error. The methodology defined the frames that produce low errors as normal frames while frames yielding high error as anomalous ones. Li et al. [7] proposed an autoencoder (AE) framework that utilized intermediate neural network layers during training to detect abnormalities in videos. A variance attention strategy was introduced to dynamically assign weights to moving objects in the video to improve the detection accuracy without counting on optical flow. The framework introduced a novel motion loss function to obtain abnormal motion directly from the original video frames. Fu et al. [8] introduced an encoder-decoder model to detect abnormalities in videos that incorporated spatiotemporal masked autoencoders (ST-MAE) with skip connections and memory modules . The model extracted high-level semantic and temporal contextual data in video activities by using spatio-temporal cubes (STCs) to characterize video activities. In addition, it used a masked autoencoder to reconstruct the masked portions of video frames, forcing the model to learn meaningful patterns. To differentiate between normal and abnormal events in videos, the model utilized memory modules to preserve normal patterns from unmasked patches across different layers. Besides, skip connections were adopted to mitigate the information loss caused by the memory modules to ensure accurate reconstruction. Lei et al. [9] proposed a technique for video anomaly detection that makes use of a Weakly Supervised Data Augmentation Network (WSDAN) and an improved U-Net architecture for multi-scale feature extraction. The method addressed the scale changes in video data by utilizing Spatial Pyramid Convolution (SPC) modules to capture visual information at several scales. By enhancing the dataset with attention-guided approaches, WSDAN is integrated to enhance training and enable more precise anomaly identification. Convolutional Block Attention Module (CBAM) is also included to dynamically modify feature map weights, reducing background noise and focusing on anomaly-relevant regions in video frames. The process of anomaly identification involved the normalization of Peak Signal-to-Noise Ratio (PSNR) between the original and forecasted frames. Ullah et al. [10] suggested a deep-learning approach that targeted the identification of abnormalities in videos from surveillance systems. The authors implemented a light-weight convolutional neural network (CNN) model to capture the spatial characteristics from video frames. Those features were then passed into a residual attention-based long short-term memory network. Thus, the model could learn temporal context information and recognize anomalous activities in surveillance videos. The goal to use residual blocks in the LSTM was to aid in saving more than 10% of learnable parameters. This was superior compared to traditional LSTM networks and help to improve the model's detection efficiency. Cheng et al. [11] proposed a reconstruction-based approach called "SecureAD " for video anomaly detection that could detect anomalies while reducing computational complexity and maintaining user privacy. The approach integrated a Conv-AE and a Conv-LSTM network. SecureAD leveraged an additive secret sharing mechanism that enabled the model to achieve privacy-preserving CNN training and detect anomalies in videos without disclosing either the raw video data or the model parameters.

III. Proposed Video Anomaly Detection Framework

Frames sequencing plays a crucial role in comprehending the video context. The context from prior frames is essential in video anomaly detection as it contributes in determining whether a particular behavior or event is anomalous. Consequently, utilizing LSTM network deals with this issue since it remembers context from previous frames. Additionally, there is a great need for quick, high accuracy, real-time systems that can identify anomalies in videos. Hence, feature extraction method is plays a role in the speed of the system..

The proposed framework, as shown in Fig. 1, is an encoder-decoder architecture, designed to identify anomalies in big datasets specifically ShanghaiTech, and is based on the recent vision transformer (ViT) architecture that captures the global context features from video frames. Additionally, this model incorporates LSTMs with an attention block called Spatiotemporal Relationships (STR-attention) to extract the spatiotemporal features of the video frames.

- *Preprocessing*: The videos are segmented into frames and each frame is then divided it into equal-sized patches. These patches are subsequently converted using linear projection into vectors to form the frame embeddings. To make sure that the framework maintains the spatial context of the frame, which is important for recognizing the image's components overall layout, the positional encodings are then appended to these patch embeddings.
- *Feature extraction:* ViT's self-attention mechanism allows focusing on different areas of the image or frame based on their contextual importance, resulting in a more comprehensive and nuanced analysis. Using the multi-head self-attention mechanism, residual connections, and layer normalization, the feature-capturing process of the vision transformer is improved. This enhances the framework's ability to identify nuanced connections and patterns while capturing contextual information both local and global.

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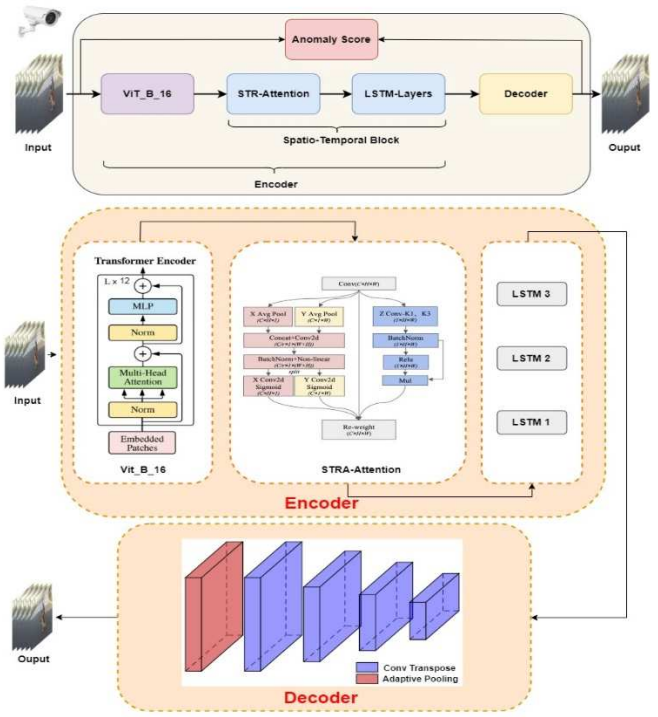


Fig. 1. The Proposed Framework Architecture

- *Operation*: The ViT-B_16 model was utilized in the proposed model as a part of the encoder for the goal of feature extraction. The ViT-B_16 draws an input frame and splits it into 16x16 patches. These patches are then linearly embedded into embedding vectors. Positional encodings are then used to maintain the spatial data included in the resulting vectors. Every patch in ViT-B_16 is encoded to a representation vector of 768 dimensions. The ViT-B_16 has 12 transformer layers, and every layer contains 12 self-attention heads. To obtain the dependencies and connections throughout the whole frame and concurrently concentrate on different regions of the input frame, the frame's patches are addressed by those multi-head self-attention and then sent to feed-forward layers. Lastly, the information from every patch is combined using the multi-layer perceptron (MLP) head. Then, an attention-based spatiotemporal model is utilized to obtain spatiotemporal connections between the video frames' objects by feeding the acquired features from the ViT to this spatiotemporal model. These characteristics are subsequently employed by the decoder part to recreate the input frames and reduce the loss of reconstruction. The decoder portion has four ConV. layers that enable quick reconstruction of the frames. An adaptive pooling layer is utilized that produced a reconstructed output frame with size equals to the input frame size.

Furthermore, the STR attention block is utilized in the proposed framework to acquire the spatiotemporal connections between video frames' objects. In contradistinction to the model introduced by Wang et al. [12], the ViT is utilized as the proposed framework's encoder part. This allows the framework to obtain the global context between the image patches besides the local features, instead of using CNN modules, which can capture local features only. To improve the video anomaly identification process, the spatiotemporal connections between objects were gathered by using the STR attention block. The STR attention block can find complex correlations and connections in video data by using both temporal and spatial attention processes. Pooling and Conv. layers are used to gather data along the vertical and horizontal axes to gain the spatial attention. The collected features are then concatenated and sent via a shared convolutional layer to focus on the significant spatial components, generating the spatial attention scores. After that, temporal pooling and convolutional procedures are used to determine connections between frames and generate the scores of the temporal attention. These spatial and temporal scores are then combined to form a spatiotemporal attention map, where the original feature maps are weighted to provide deeper attention to regions with significant spatiotemporal correlations. The operation details of the STR-attention model are illustrated in the next paragraphs.

Two pooling kernels are used in a one-to-one dimensional feature encoding process to encode each channel along the vertical and horizontal axes of the input $X \in \mathbb{R}^{(C\times H\times W)}$. These kernels have spatial sizes of (H,1) and (W,1). The output of the C-th channel in the H dimension is calculated as following:

$$Y_c^h(h) = \frac{1}{W}\sum_{0\le i<W} x_c(h,i) \tag{1}$$

where the input X, which normally originates from the fixed kernel size convolution layer, represents the characteristics in the H dimension. In addition, the C-th channel output of the W dimension is calculated using the formula below:

$$Y_c^w(w) = \frac{1}{H}\sum_{0\le j<H} x_c(j,w) \tag{2}$$

Then, the characteristics of the sequence of the video frame at the time t are obtained from every channel feature as follows:

$$Y_t(X) = F^{3x3}\sigma\left(F^{1x1}(X)\right), X \in \mathbb{R}^{C/r\times(H+W)} \tag{3}$$

where F represents the convolution operation and σ represents the normalization procedure. The reduction ratio, r, which is a controller for the number of channels' variation. The convolution kernel sizes are indicated by 3 × 3 and 1 x 1. The results of the three processes discussed above are gathered along each of the three dimensions to create a collection of spatiotemporal-aware feature mappings. This aggregation operation is carried out by adding the feature maps acquired from Equations (1) and (2) as follows:

$$Y^{hw} = \delta(F^{1x1}[Y^h, Y^w]) \tag{4}$$

where δ stands for the nonlinear activation function, F1×1 denotes the convolutional transformation via a convolutional kernel of size 1 × 1, and [.,.] denotes the concatenation of two tensors.

The output Y is then divided into two tensors, $Y^h \in \mathbb{R}^{C/r\times H}$ and $Y^w \in \mathbb{R}^{C/r\times W}$, that have the same dimension. The activation function is then applied to each of the split tensors in the following manner to produce the spatial and temporal attentions:

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$$g^h = \text{Sigmoid}(F_h(Y^h)) \quad (5)$$

$$g^w = \text{Sigmoid}(F_w(Y^w)) \quad (6)$$

where the two distinct 1 x 1 convolutional kernels are indicated by F_h and F_w. The area feature weight tensor g_t that is sensitive to time change is then represented by $Y_t(X)$, which is produced by the nonlinear activation function as follows:

$$g^t = \text{Relu}(Y_t(X)) \quad (7)$$

Finally, the attention weighted output Y is calculated as follows:

$$Y_c(\text{i},\text{j}) = X_c(\text{i},\text{j}) \times g_c^h(i) \times g_c^w(i) \times g_c^t(i,j) \quad (8)$$

As a result, the forementioned three operations provide the long-range temporal and spatial connections, allowing the framework to precisely identify the spatiotemporal links among objects. Consequently, by utilizing temporal interdependence and spatial linkages between frames' objects, the attention mechanism could capture the region of interest.

LSTM layers enhance the model's ability to remember and process long-term dependencies between video frames. Hence, three LSTM layers are used to maintain the internal state that enables the model to remember previous inputs. This context-aware learning allows the model to acquire normal patterns more consistently with the temporal changes in events, thereby increasing the overall accuracy of identifying anomalies. LSTMs can model the long-term dependencies between frames, which enhances the STRA-attention block's capacity to effectively record spatial and short-term temporal interactions. Together, these elements allow the model to identify abnormalities that are concentrated in one area and those that spread over a longer video sequence.

Four CNN layers constituted the decoder that receives the spatiotemporal features extracted from encoder part (ViT + spatiotemporal block) to reconstruct the input video frames. By producing frames that closely to the original input ones, this reconstruction approach seeks to minimize the reconstruction loss. An adaptive pooling layer is included, to guarantee that the output frames have the same dimensions as the input. This enables the model to discriminate normal from anomalous frames depending on the reconstruction error.

In our work, the reconstruction error is used as the anomaly score. The mean squared error (MSE) is utilized as the reconstruction loss by calculating the mean squared differences between the input and reconstructed frames to assess how differ the reconstructed frame from the input one. As the goal from training the proposed model is to reconstruct well the normal video frames. Consequently, the obtained values of the MSE are low for regular frames and high for anomalous ones since the model is trained on normal frames. Thus, robust optimization is utilized to reduce the reconstruction loss between the input ground truth Y and reconstructed output $\hat{Y}$ by the decoder. The reconstruction error is calculated as follows:

$$L_{rec} = \left\| \hat{Y} - Y \right\|_2 \quad (9)$$

IV. Experimental Results And Discussion

A. Datasets

The experiments in this work are conducted on three prominent public benchmark video datasets widely recognized in the field of video anomaly detection, UCSD Ped2, CUHK Avenue, and ShanghaiTech. Fig.2 shows different normal and anomaly samples from the three datasets and Table I shows the utilized datasets summarization.

UCSD-Pedestrian 2 (Ped2) [13]: one of the most known video anomaly detection datasets. The dataset contains records for normal pedestrian activities and rare abnormalities in a controlled environment. The videos represent a single scene and were recorded at different times during the day. The normal events include people moving along walkways, while abnormalities include strange appearance and motion objects such as carts, bicycles, skateboards, and standing on grass, etc. Ped2 contains 16 training videos and 12 testing videos with 2550 frames and 2010 frames, respectively. The frames' resolution is 240 by 360 pixels.



Fig. 2. Examples of normal and anomaly events from the UCSD, Avenue, and ShanghaiTech datasets

The CUHK Avenue [14]: offers a wider range of irregularities in a public university environment. The films are taken from the campus avenue, which represent a single scene. The dataset contains 15,324 testing frames and 15,328 training frames, divided across 21 test videos and 16 training videos. About 47 abnormal behaviors, including loitering, throwing things, and racing through the fence, were recorded in this dataset. Each frame has a resolution of 360 by 640 pixels. The training and testing videos in the CUHK Avenue dataset are less than two minutes.

ShanghaiTech [15]: is considered the largest and most comprehensive unsupervised dataset currently accessible for video anomaly identification. It has 437 movies from 13 cameras spread out over the ShanghaiTech campus at an 856 by 480-pixel frame resolution. It includes 330 training movies with 274,515 frames of only normal events and 107 test videos with both normal and 130 abnormal events. It has various videos with challenging lighting, and unique camera perspectives. Activities including skateboarding, riding motorbikes and bikes, pursuing, fighting, and running are among the human oddities in the dataset. Every video has annotations for frames and pixels.

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TABLE I. SUMMARY OF VIDEO ANOMALY DATASETS

Dataset	#Training Frames \ Videos	#Test Frames \ Videos	Anomaly Types
UCSD-Ped2	2550\16	2010\12	• Skateboard Bicycles • Carts, etc.
CUHK Avenue	15,328\16	15,324\21	• Throwing • Loitering • Jumping • Running, etc.
Shanghai Tech	274,515\330	40,791\107	• Bikes • Chasing • Fighting • Robbing • Jogging, etc.

The datasets used in this work show different challenges that experience the robustness of the proposed model. For example, the ShanghaiTech dataset includes scenes with significant lighting variations, diverse camera angles, and a set of anomalies such as truck, van, biking, running, kicking, falling, and fighting, captured from 13 different cameras. Similarly, CUHK Avenue has irregularities such as throwing objects, picking up objects, loitering, walking in unusual directions (e.g., wrong way or against the usual flow), and rapid movement, filmed in a public university setting. In addition, there is a slight camera shake in some videos such as in test video 2 and some normal patterns seldom appear in training data. The UCSD Ped2 dataset, though more controlled, includes anomalies like skateboarding and bicycling that contrast with regular pedestrian behavior. Besides, some anomaly behaviors and objects are distant from the camera. These conditions simulate the complexities encountered in real-world surveillance, which the proposed ViT-based framework is designed to manage effectively.

B. Experimental Setup

Following the settings in [3], the proposed framework was trained using Adam optimizer [16]. The learning rate, the batch size and the input sequence length are $1e^{-4}$, 16, and five frames, respectively. The implementation of the proposed model was performed a 16 GB of memory NVIDIA GeForce RTX 3080 GPU, using Pytorch backend and Python 3. The extracted frame from each video was resized to 224 × 224 × 3 and passed as input to the framework. The area under the receiver operating characteristic (AUC ROC) curve was used to evaluate the model's performance in accordance with previous studies, to assess how well the proposed approach could differentiate between abnormal and normal frames.

C. Performance Evaluation

To assess the anomaly detection performance on the three datasets UCSD-Ped2, CUHK Avenue, and ShanghaiTech. We conducted our experiments using ViT model without the patio-temporal STR-attention + LSTM block and considered it as the baseline and then with this block added to the ViT . The obtained results are shown in Table II. From the table, it is obvious that using STR-attention + LSTM block enhanced the results for the large heterogeneous ShanghaiTech dataset, which improves the capability of the introduced framework to identify anomalies in videos taken under different environmental and lighting circumstances.

Fig.3 and Fig.4 show two test videos from UCSD-Ped2 dataset. The figures depict how the model can respond successfully to the anomaly events (e.g. cyclist and skater) with high accuracy scores against different anomaly.

TABLE II. AUC ROCS (%) FOR THE UCSD, AVENUE, AND SHANGHAITECH DATASETS.

Model\Dataset	UCSD%	Avenue%	Shanghai%
Baseline	94.5	84.9	77.3
Baseline + STR-attention + LSTM	93.8	83.4	79.6

In the same way, Fig. 5 and Fig.6 depict the anomaly scores for test videos 05 and 06 from Avenue dataset e.g. (playing with bag and wrong way direction), the proposed model can effectively detect anomaly events even if the anomaly event or object remains for a short period in the scene, as in Test video 06, which has two short periods of anomaly events.

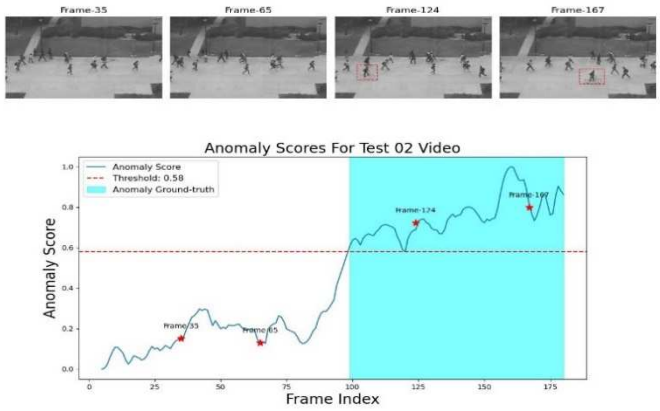


Fig. 3. UCSD-Ped2 Test 02 Video Anomaly score curve (anomaly type: cyclist)

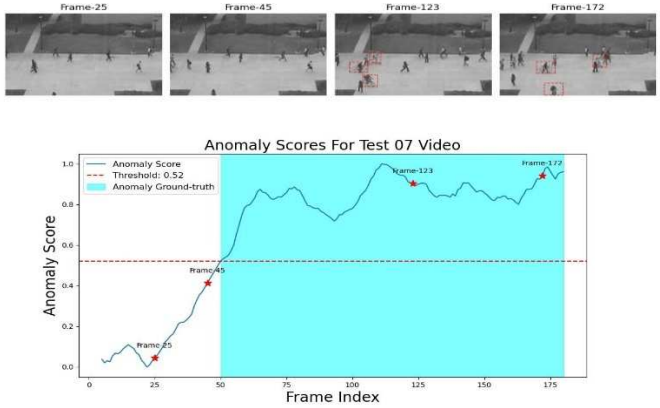


Fig. 4. UCSD-Ped2 Test 07 Video Anomaly score curve (anomaly type: cyclists + Skater)

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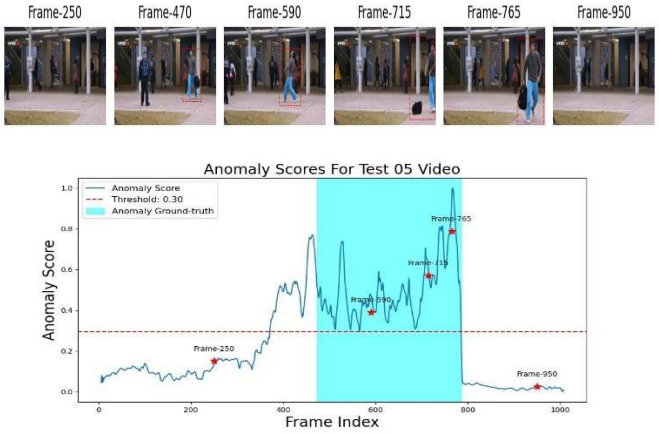


Fig. 5. Avenue Test 05 Video Anomaly score curve (anomaly type: Bag throwing)

For the ShanghaiTech dataset, which contains videos from 13 different cameras, two videos from two different scenes are displayed in Fig.7 and Fig.8. The first one shows a person running in front of the camera. While the second one shows a person falling on the ground. The model shows outstanding ability in the detection of anomalies for this large multi-scene dataset.

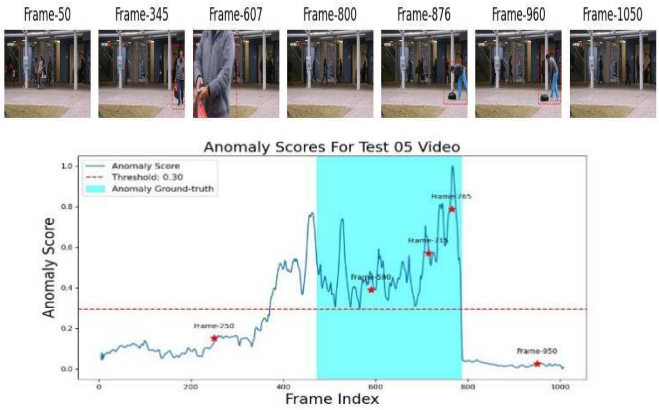


Fig. 6. Avenue Test 06 Video Anomaly score curve (anomaly type: Bag throwing + Wrong way walking)

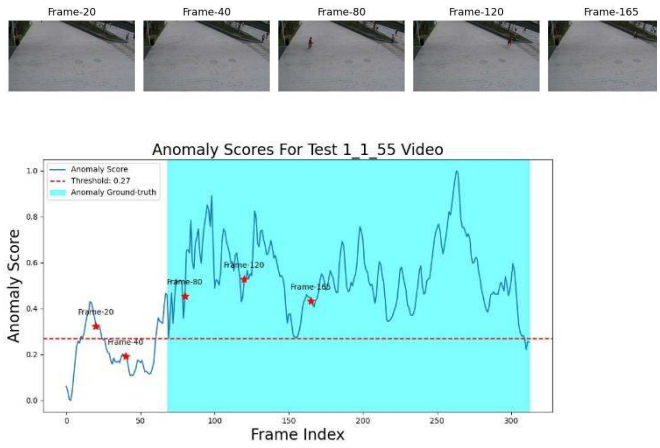


Fig. 7. Shanghia Test 1_1_155 Video Anomaly score curve (anomaly type: Running)

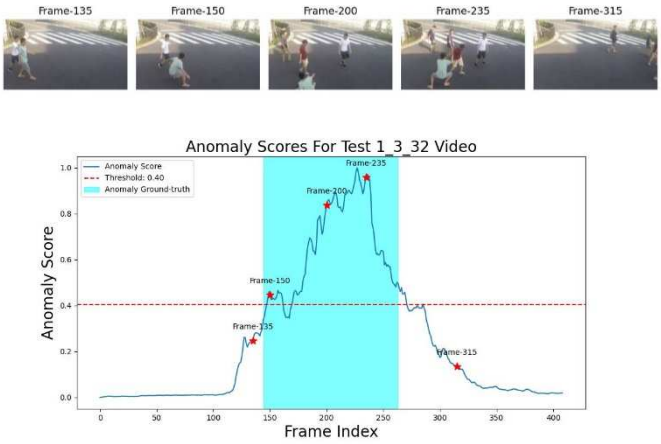


Fig. 8. Shanghia Test 1_3_32 Video Anomaly score curve (anomaly type: Falling)

To summarize the obtained results, the combining of the ViT with the spatio-temporal attention mechanism and the LSTM layers showed the ability of the model to learn intricate patterns and extract discriminative features to detect anomalies with high anomaly scores for different datasets that represent distinct environments, achieving an outstanding AUC_ROC of 79.6% for the largest shanghai dataset. The proposed model and the obtained results put a step forward for introducing a real-time video anomaly detection systems for surveillance systems.

To show the performance of the proposed framework, the results of the introduced framework is compared to state-of-the-art, as shown in Table III, that depicts the efficiency of the model for the large heterogeneous datasets.

TABLE III. PERFORMANCE COMPARISON OF AUC ROCS (%)WITH STATE-OF-THE-ART ANOMALY DETECTION METHODS.

Reference	Model	UCSD-Ped2	Avenue	Shanghai-Tech
[17]	Memory-guided AE	97.0	88.5	70.5
[18]	Attention residual AE	97.4	86.7	73.6
[19]	Memory-augmented AE	94.1	83.3	71.2
[20]	AE + DPU + Attention module	97.9	85.9	-
[21]	Gaussian Mixture Fully Convolutional Variational Autoencoder (GMFC-VAE)	92.2	83.4	-
[22]	AE + Probability distribution density estimator	95.4	-	72.5
[23]	Two-stream clustering Autoencoder	96.5	86.0	73.3
[24]	Residual autoencoder	83.0	82.0	-
[25]	Two-stream autoencoder	84.5	80.3	-
[26]	Attention-based ConvLSTM network þ Conv Autoencoder	95.6	-	73.1
[27]	Noise AE + AE + Second-order channel attention	97.9	86.2	-

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[28]	Residual AE	95.4	80.9	-
[29]	ConvLSTM-AE	97.9	89.8	73.7
[30]	Implicit two-path AE + normalizing flow	97.3	85.8	74.7
[31]	Transformer memory + AE	98.1	88.5	72.5
[9]	Multi-scale feature attention U-Net + weakly supervised data augmentation	97.9	86.2	-
Ours	ViT + STR-attention + LSTMs	93.8	83.4	79.6

The proposed ViT-based anomaly detection model may face difficulties with constrained computational resources in practical surveillance systems. The substantial computational requirements of the Vision Transformer (ViT) model may need hardware acceleration (e.g., GPUs or TPUs) to sustain real-time processing efficiency. Moreover, the reliability of anomaly detection in real surveillance videos may be affected by the differences in video quality, camera perspectives, and lighting conditions. On the other hand, the three datasets utilized in training and testing the proposed model represent different environments that simulate the circumstances in real surveillance videos. This may indicate the reliability of the proposed framework if applied in real-time surveillance situations. To assess the computational complexity of the proposed model, the number of floating-point operations (Flops) and parameters of the model could be calculated. These two parameters are independent of hardware and frameworks rather than the inference time per frame, which depends on the utilized resources, e.g., CPUs or GPUs. The proposed model has 165.66 G Flops and 260.25 M parameters.

V. Conclusion

A novel approach for video anomaly detection in large datasets using the ViT, STR-attention mechanism, and LSTMs, is proposed in this work. In surveillance systems and their applications, the detection accuracy of the anomalies is crucial and the proposed model significantly improves the anomaly detection performance, achieving state-of-the-art results. The recent ViT was incorporated in this work to capture global context characteristics and relationships within the video frame. The STR-attention block was incorporated to get the spatiotemporal features of the input frame sequence, besides the relations within the frames' objects, with the aid of LSTMs to improve the context-aware learning of the model and enhance the anomaly detection overall accuracy. The challenge associated with the large and variations in the datasets that captured in different environments with different environmental and lighting conditions, is effectively addressed, resulting in an enhanced and robust video anomaly detection model in for detecting anomalies in surveillance systems. The MSE was used for the anomaly assessment process to accurately determine whether any anomalies were present in the video frames. The experimental findings, which were obtained using the UCSD Ped2, CUHK Avenue, and ShanghaiTech datasets, provide strong evidence for the superiority and efficacy of this method for video anomaly identification, especially for large datasets, which the model is primarily designed to tackle.

It should be highlighted, however, that there are still issues with this study, including high computing complexity, a lot of parameters, and restricted scene perception. Thus, Future studies will focus on improving this method's performance and applicability, yielding significant findings for video surveillance research. Although the suggested approach performs well on the utilized benchmark datasets, ViT's processing cost might make real-time deployment difficult, particularly in systems with limited resources. Hence, more investigations could be carried out in to reduce the computational demands for real-time applicability. Moreover, we use reconstruction error as the anomaly scoring mechanism, which is standard in video anomaly detection. However, future research could explore other anomaly-scoring techniques that might better leverage the features learned by the Vision Transformer (ViT) and STR-attention modules. For instance, feature-based scoring methods, such as clustering of high-dimensional ViT feature embeddings or distance metrics on attention-weighted features, could be investigated. This could be useful in allowing more nuanced differentiation between normal and anomalous events by concentrating on obtained spatiotemporal representations instead of the reconstruction loss alone.

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